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COMP 3008: Human-Computer Interaction
Assignment 3

1.

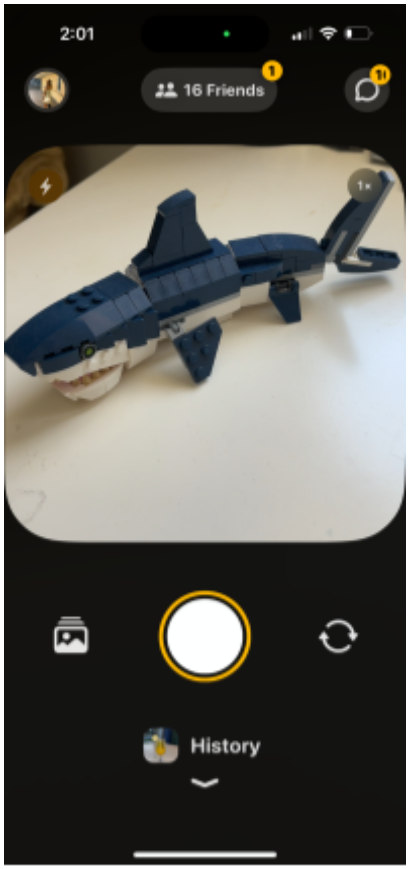

Figure 1: Locket Homescreen

a. Locket Widget App

One task a user might do with this interface is send a photo to a friend. The user opens the Locket widget on their home screen, a unique feature that sets it apart from standard app icons (Figure 2). Then the user taps it to activate the camera, takes a quick selfie or snaps a photo of what they're doing, adds a small emoji or short caption, and hits send. Within seconds, the photo appears directly on their friends' home screens, allowing users to share moments instantly and stay connected.

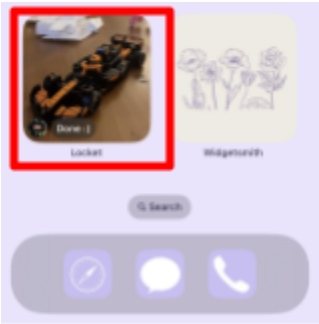

Figure 2: Widget of Locket on home screen of Iphone

b.

i.

The main goal of my interviews is to understand users' thoughts, emotions, and decision making processes that occurred during the observation tasks. While the think-aloud observations showed what users did on the Locket app, the interviews helped uncover why they acted in certain ways, such as what caused hesitation, confusion, or satisfaction. By asking users to reflect on their experiences afterward, I hope to gather qualitative data that complements the behavioral insights from the observations. This step will strengthen my overall usability evaluation by connecting user actions to their perceptions and expectations, building on my three core usability goals: effectiveness, learnability, and enjoyability.

In connection to **Goal 1: effectiveness**, my interviews will include questions about how confident users felt when sending or receiving photos through the widget and whether they believed their actions were successful. I plan to ask about any points of confusion or delay they noticed, such as unclear messages or feedback from the system. These questions will help determine whether the app communicates effectively and allows users to complete its main purpose to photo share clearly. Understanding how users describe these experiences in their own words will also help me identify specific points where the app fails to provide sufficient feedback or clarity.

The interviews will also address **Goal 2: learnability** and **Goal 3: enjoyability** by exploring users' emotional and cognitive experiences with the app. I will ask them to describe what felt intuitive or confusing when learning how to use features such as adding friends, deleting photos, or reacting to messages, as well as how enjoyable or frustrating the overall experience was. These reflections will help reveal whether the app supports new users effectively and whether it creates the positive, socially connected experience it aims for. By combining these insights with the earlier observations, I will be able to form a more complete picture of how users understand, learn, and emotionally respond to the Locket app.

ii. Interview Questions:

Can you describe your overall experience using the Locket app so far?

Was there anything that confused or frustrated you while trying to send or view a photo on the app?

How easy or difficult was it to learn how to use the app the first time you tried it?

When using the app, did you feel confident that your actions (like sending a photo or adding a friend) were successful?

Would you recommend this app to a friend? Why or why not?

iii. For this study, a semi-structured interview approach will be used. This method allows for flexibility while maintaining a focus on the main usability goals. Semi-structured interviews are ideal for following up after user observations because they allow the interviewer to ask planned questions related to specific tasks while also giving participants the freedom to elaborate on their experiences. This structure supports both comparability across interviews and deeper insights into user thoughts, feelings, and challenges.

The recording method will include using an audio recorder on the Voice Memos app to capture the full conversation, along with written field notes taken during the session. Audio recording ensures that no important details or quotes are missed and allows for accurate transcription and analysis afterward. Field notes will complement the recording by capturing non-verbal cues, tone, or immediate reflections that might not be evident in the audio alone. This combination enhances reliability of the data and goes further in depth. Furthermore, the location of the interview will be where the users prefer for convenience and comfort. A comfortable participant is more likely to provide open and honest feedback.

Semi-structured interviews allow flexibility to probe when users mention unexpected issues, while recordings ensure that all details are preserved for later coding and thematic analysis. The use

of both recording and notes also reduces the risk of losing critical information or user feedback, making the results more valid and comprehensive.

iv.

User 1 Transcript of Interview
Setting: User's home, at the dinner table **Interviewer Notes:** User was calm and thoughtful. They often paused before answering and made a few confused facial expressions when discussing the setup process. **Interviewer**: "Thanks for your time, [user's name]. To start, can you describe your overall experience using the Locket app so far?" **User 1:** "Hmm, overall? It's... a fun idea I think. I like the concept of having a little photo widget from a friend right on my home screen, cause I get to see them whenever I open my phone. That part feels personal. But the actual process of getting it set up was lowkey harder than I thought. I felt like it could've been easier, you know?" **Interviewer**: "You mentioned the setup is difficult. Can you tell me more about what confused or frustrated you while trying to send or view a photo?" **User 1:** "Yeah, definitely. The whole 'adding a friend' part was the big one. I wasn't sure if I was supposed to share a link, or if we both needed to have usernames, or what. The app didn't give much guidance. Then, when I sent my first photo, it just... disappeared into the void. I didn't get any confirmation it was sent, and my friend didn't get it for like, ten minutes. I ended up just texting them to check, which kind of defeated the purpose." **Interviewer**: "So, a lack of guidance and slow or missing confirmations. Based on that, how easy or difficult was it to learn how to use the app the first time you tried it?" **User 1**: "I'd say it was moderately difficult. It's not super complex, but it makes you figure things out on your own. Taking a photo and hitting send are easy. But the stuff around the edges, like managing the widget and understanding the friend system, has a bit of a learning curve. I didn't feel like the app taught me how to use it very well." **Interviewer**: "That leads well into my next question. When using the app, did you feel confident that your actions, like sending that photo or adding a friend, were successful?" **User 1**: "Not really, no. Especially with sending photos. Like I said, there's no 'whoosh' sound or a confirmation. You just hope it worked. With adding a friend, I only knew it was successful when they suddenly appeared in my list. There was no notification like, 'Your friend has been added!' So, no, I lacked confidence I guess." **Interviewer**: "Finally, would you recommend this app to a friend? Why or why not?"

User 1: "Right now? Probably not. I might tell them the idea is cool, but I'd warn them that the setup is iffy and you have to be patient. If they fixed the confirmation issues and made adding friends more easier , then yes, absolutely. But in its current state, it feels a bit too unreliable to recommend not going to lie."

Interviewer: "Thank you so much for your time!"

User 2 Transcript of Interview

Setting: At users house, in the basement couch
Interviewer Notes: User was energetic and spoke quickly. They frequently used hand gestures and smiled when talking about seeing their friend's photo.

Interviewer: "Hi [User's name], thanks for meeting me. To begin, can you describe your overall experience using the Locket app so far?"

User 2: "Yeah, it's been fun! It's a cute app. My best friend and I downloaded it together when we went to different universities to try and stay connected, and it's a nice surprise to see a new picture of her dog or her coffee pop up on my screen during the day. It's just a sweet, low pressure way to stay in touch."

Interviewer: "That's great to hear. Was there anything at all that confused or frustrated you, even a small thing, while trying to send or view a photo?"

User 2: "Okay, yes, one thing was weird. I accidentally sent a photo to the wrong person once. The interface for selecting which friend to send to wasn't very clear, the names are small and I just tapped the wrong one. There's no 'undo send' or anything, so I just had to send a follow up message like 'Oops, that was for Sarah!' It was a little awkward."

Interviewer: "That's an interesting point we haven't heard before, an error in selecting the recipient. So, thinking back, how easy or difficult was it to learn how to use the app the first time you tried it?"

User 2: "The very basics were easy. Taking a picture and sending it is straightforward. They had instructions on how to set everything up, so pretty easy I would say.

Interviewer: "Aside from the wrong person incident, did you generally feel confident that your actions were successful? For example, when you send a photo to the right person."

User 2: "Mostly, yes. But only because my friend would usually react to it or mention it later. The app itself doesn't do a great job of making you feel confident. It doesn't tell you 'Photo Seen' or anything. You're just trusting that it went through. So my confidence came from my friend's response, not from the app."

Interviewer: "And lastly, would you recommend this app to a friend? Why or why not?"

User 2: "You know, I actually already have! I recommended it to my cousin. She loves it and will not stop sending me photos. [Laughs] So I'd recommend it for the fun and connection, but I must say, there is definitely room for improvement."

Interviewer: "Great, thank you so much for your time!"

c.

i. This analysis uses a deductive approach. The unit of analysis is a phrase within the interview response that expresses a complete thought related to the usability goals. The three primary codes were selected directly from the core usability goals stated in the project brief. The first one is effectiveness, applied to statements about successfully completing core tasks (e.g., sending a photo, adding a friend) and the app providing clear, timely feedback. Includes both positive (successful) and negative (unsuccessful, unclear) experiences. The second one is learnability: applied to statements about the initial experience of learning to use the app, including the intuitiveness of the interface, the clarity of setup processes, and the need for external help or guidance. The last one is enjoyability, applied to statements about the emotional, hedonic, and social experience of using the app, including feelings of fun, frustration, connection, and overall satisfaction.

ii.

Coding for Question - "Was there anything that confused or frustrated you while trying to send or view a photo on the app?"		
Users	**Response**	**Codes**
User 1	"Yeah, definitely. The whole 'adding a friend' part was the big one. I wasn't sure if I was supposed to share a link, or if we both needed to have usernames, or what. The app didn't give much guidance. Then, when I sent my first photo, it just... disappeared into the void. I didn't get any confirmation it was sent, and my friend didn't get it for like, ten minutes. I ended up just texting them to check, which kind of defeated the purpose"	Colour 1: learnability Colour 2: effectiveness
User 2	"Okay, yes, one thing was weird. I accidentally sent a photo to the wrong person once. The interface for selecting which friend to send to wasn't very clear,	Colour 1: Effectiveness Colour 2: Enjoyability

	the names are small and I just tapped the wrong one. There's no 'undo send' or anything, so I just had to send a follow up message like 'Oops, that was for Sarah!' It was a little awkward."	

Coding for Question - “Would you recommend this app to a friend? Why or why not?"		
Users	**Response**	**Codes**
User 1	"Right now? Probably not. I might tell them the idea is cool, but I'd warn them that the setup is iffy and you have to be patient. If they fixed the confirmation issues and made adding friends more easier , then yes, absolutely. But in its current state, it feels a bit too unreliable to recommend not going to lie."	Colour 1: Learnability Colour 2: enjoyability Colour 3: effectiveness
User 2	"You know, I actually already have! I recommended it to my cousin. She loves it and will not stop sending me photos. [Laughs] So I'd recommend it for the fun and connection, but I must say, there is definitely room for improvement."	Colour 1: effectiveness Colour 2: Enjoyability

iii.

The interviews provided important insight into the app's effectiveness, revealing a significant gap in system feedback that directly undermines user confidence. While observations may have shown users successfully sending photos, the interviews uncovered they feel hesitant because of the lack of trust in the process. Users described actions like a photo "disappear[ing] into the void" and reported having to use external communication to verify success, highlighting that the app fails to provide clear confirmation. This core failure in communication means users cannot be sure their primary goal, sharing a photo has been accomplished, making the app feel unreliable despite its functional purpose.

Furthermore, the analysis uncovered major learnability issues, particularly with the initial setup. The interviews moved beyond simply observing that users struggled with the widget

or adding friends and explained the cognitive roadblocks they faced. Participants articulated specific points of confusion, such as not understanding the mechanics of adding a friend. This demonstrates that the app does not effectively onboard users, creating a steep initial barrier.

Ultimately, the interviews revealed a fragile sense of enjoyability, one that is entirely dependent on the social outcome rather than the experience of using the app itself. Users consistently expressed that they found the concept fun and that it made them feel connected to friends. However, this positive emotional response was consistently qualified and diminished by the frustrations stemming from the app's usability flaws. The result is a conflicted user who enjoys the social connection but is hesitant to recommend the app, proving that while the app's idea is good but the functional shortcomings prevent it from delivering a consistently positive and recommendable experience.

2.

a. The goal of the survey is to gather quantitative feedback on how users perceive the usability and overall experience of using the Locket app, focusing on effectiveness, learnability, and enjoyability, the same key usability goals I established during the observation and interview stages. By collecting numerical data, the survey will help identify patterns in user satisfaction and provide measurable evidence of how well the app meets its intended purpose. Specifically, the survey aims to determine whether users find the app easy to use and understand (learnability), whether they can successfully complete main tasks like sending and receiving photos without issues (effectiveness), and whether the app feels engaging and emotionally positive to use (enjoyability).

I am planning on using the System Usability Scale (SUS) because it effectively captures overall usability, including effectiveness, efficiency, and satisfaction. These aspects align directly with the app’s usability goals. For example, SUS items assess how easy and intuitive the system is to use, which connects to learnability and effectiveness, while overall satisfaction relates closely to the enjoyability goal. Using SUS provides standardized and validated metrics that can be compared across participants, ensuring that the results are reliable and interpretable.

Alongside the SUS, I am creating a questionnaire tailored to the Locket app that will be used to capture more task specific usability feedback. These closed ended questions will focus on aspects such as ease of setup, clarity of feedback messages, and satisfaction with photo-sharing features. Combining SUS with the custom questionnaire allows for both a general usability assessment and focused insight into the unique features of the app. This mixed survey approach ensures a comprehensive understanding of how well the interface supports users’ needs and emotional engagement.

b. https://forms.gle/QxUK1pvQywiKocqWA

c.

The first two questions are to get key demographic information from users.

Question	**User 1 initial**	**User 1 Reversed**	**User 2 initial**	**User 2 Reversed**
3	4	4	4	4
4	2	4	3	3
5	4	4	2	2
6	5	5	3	3
7	4	2	2	4
8	4	4	5	5
9	4	4	4	4
10	5	5	4	4
11	4	2	4	2
12	5	5	3	3

Table 1: Answers and reversed coded answers of the Users from questionnaire.

Users	**SUS Final Value**
User 1	70
User 2	60
Average	65

Table 2: Final values of the System Usability Scale Questionnaire for each user

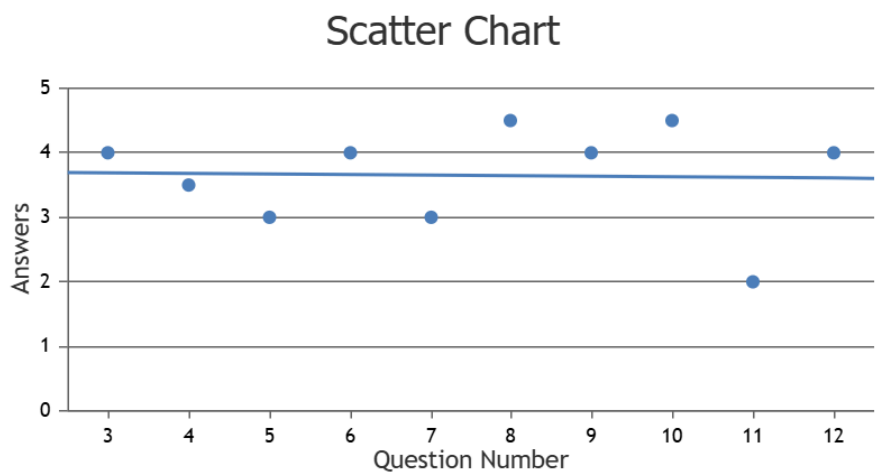

Chart 1: Scatter plot chart for average aggregate results

d. The results from my survey and the SUS survey shows that people really like the main idea behind the Locket app. We learned that the app's main strength is its enjoyability. Users reported that sharing photos was fun and made them feel more connected to their friends. This shows the app's social purpose is successfully connecting with people on an emotional level. However, the survey also uncovered significant problems in other areas. The app received a usability average score of 65. Since the common benchmark for a "good" app is 68, our score of 65 tells us the app is just "okay," but has clear room for improvement. This average score is pulled down mostly by issues with learnability and effectiveness. Users directly stated that the app was difficult to learn at first, especially for basic tasks like adding friends. This is a major learnability problem. Furthermore, there were effectiveness issues where users were not always confident their actions worked because the app failed to give clear confirmation, like showing if a photo was successfully delivered.

The scatter chart shows that user responses remain relatively consistent across questions, with most answers clustering between 3 and 4, indicating generally neutral to positive feedback. The trendline is nearly flat, suggesting there is no strong upward or downward trend in user satisfaction or performance across the questions. However, Question 11 stands out with a noticeably lower score around 2, which may point to a specific issue or negative user experience related to that task. Overall, the data suggests that users found most aspects of the app moderately effective and satisfactory, but certain areas may require further attention to improve consistency in user satisfaction.

In summary, the survey shows that users find the Locket app enjoyable in its purpose, which is a great foundation. But this positive feeling is weakened because the app is difficult to learn and doesn't always feel reliable to use. To improve, we don't need to change the fun concept. Instead, we need to focus on making the initial setup simpler to improve learnability and

adding better feedback to strengthen effectiveness. Fixing these issues would make the app easier and more trustworthy to use, which would ultimately make the enjoyable experience even better for everyone.

3.

a.

Persona 1:

- **Name**: Alexandra Chen
- **Background & Experience**:
    - Alex is a high school student who is tech-literate but cautious with new apps. She values cute aesthetic design with effectiveness. Shuses messaging apps like WhatsApp and Imessage daily for their clarity and reliability, and is wary of apps that feel gimmicky or waste her time. She downloaded Locket Widget to feel closer to her long-distance partner without the pressure of constant texting.
- **Needs & Frustrations:**
    - Needs: A reliable and predictable way to share small moments to her partner and clear confirmation that actions have been successful, an intuitive setup that doesn't require a tutorial.
    - Frustrations: She is deeply frustrated by ambiguity. The lack of a "sent/delivered" confirmation in the app makes her anxious. She finds the process of adding a friend and setting up the widget confusing and poorly explained.
- **Motivation & Goals:**
    - Motivation: To maintain a lightweight, low-pressure, but consistent connection with her partner. She is motivated by the idea of a shared, ambient space on his phone to see his partner and his everyday life.
    - Goals: To complete the setup in one try without external help, to feel 100% confident that a photo has been sent and received and to integrate the app seamlessly into his daily routine without it causing extra hassle or doubt.

Persona 2:

- **Name**: Sam Rivera
- **Background & Experience:**
    - Sam is a university undergraduate student who works in a creative field and is an early adopter of social and visual apps like Instagram and BeReal. They are enthusiastic about new ways to communicate and are the person in their friend group who finds and shares new apps. They value fun, visual expression, and novelty over technical perfection.

- **Needs & Frustrations:**

- Needs: A fun and engaging way to surprise friends, features that support spontaneous sharing and an app that feels personal and visually cohesive.
- Frustrations: Sam is frustrated by design flaws that interrupt a fun flow. They were annoyed by the unclear interface that led to sending a photo to the wrong person and the lack of an "undo" option.

- **Motivation & Goals:**
  - Motivation: To create a sense of playful, ongoing connection with their close-knit circle of friends. They are motivated by the joy of the surprise and the reaction it gets.
  - Goals: To quickly onboard friends into the app without having to give tech support, to share moments with the correct person effortlessly and to be able to recommend the app to others and feel 100 percent confident that their recommendation is good

b.

Starting assumptions:

- You already have Locket App installed and an account created
- You have at least one friend already added to their friend list in the app
- Start: You are on your home screen with the locket widget.
- End: After you sent a photo
- Level of detail: Clicks

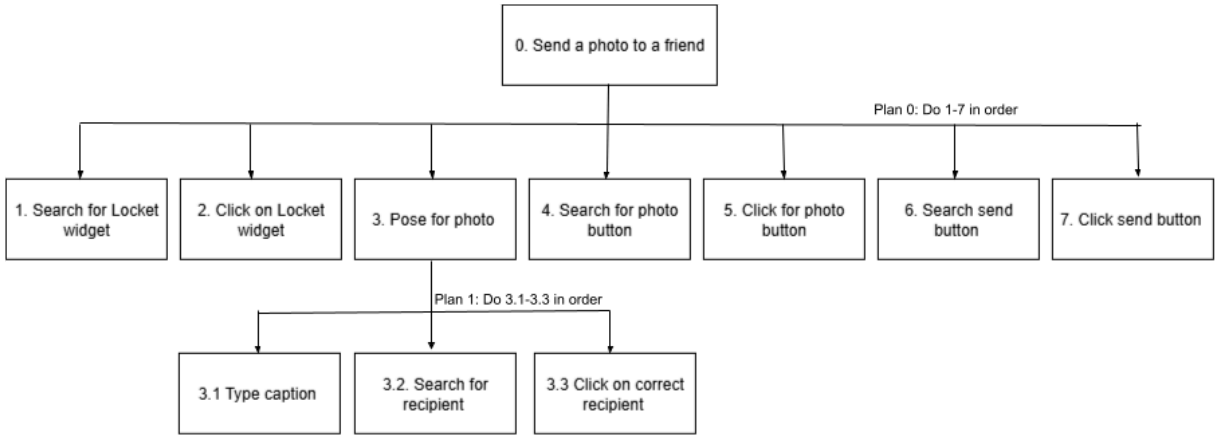

Figure 3: Cognitive Task Analysis for Sending a photo to a friend

c.

A relevant conclusion from these personas is that the app's fundamental design must prioritize error prevention and instant, understandable feedback to build essential trust with every user. This insight directly guides the design phase by focusing our efforts on creating safeguards and clear communication, for example, we will design the "send" interface to prominently display the recipient's name to prevent mis-sends, and we will implement a simple, always-visible status icon, like a checkmark to immediately confirm a photo was successfully sent and delivered. By embedding this clarity and reliability into the core user actions, we directly solve Alexandra's anxiety about whether her connection is active and eliminate Sam's frustration with accidental mistakes, thereby creating a foundation that feels both secure for both personas.

The CTA gives us a flowchart and process of sending a photo with the app. It shows that the process is a very linear process. It is a simple task, however it has many steps. This conclusion may help for the redesign process, for example, to streamline and consolidate the workflow. Our goal will be to reduce the cognitive load and number of decisions required from the user. Instead of a 7-step plan, we will design for a primary flow that can be accomplished in 2-3 intuitive steps. For example, tapping the widget could open the camera immediately, and releasing the shutter could send the photo automatically to a default recipient. This shift towards a simplified, "one-tap" philosophy would be efficient and less work for the user. In conclusion, these insights will help moving forward in the design phase by ensuring that future iterations of the app are both trustworthy and effortless to use, allowing users to feel more confident and connected while completing core tasks with minimal effort.

4.

a. How useful were the techniques and approaches used here (interview, survey, CTA, personas)?

The interviews and surveys were very helpful tools that were used in this assignment. This was helpful because I got a clear, direct message from the people who use the app. The surveys were great because they helped us count how many people were having the same problems, like not knowing if their photo was sent. The interviews were even better because they let us ask "why?" We could understand the feelings behind the problems, like how a missing "sent" notification made someone feel anxious instead of just annoyed. This information was our most valuable guide. This quantitative and qualitative feedback served as a concrete foundation, revealing what users actually do and what they explicitly say their problems are. This raw data is invaluable as it gives us a solid, trustworthy list of the biggest issues that we needed to fix to make users happy.

On the other hand, the personas and the CTA weren't very helpful in this context, not because the methods are flawed, but because of how they were applied. The personas felt like we were creating fictional characters based on what we already knew, instead of discovering new insights. They didn't tell us anything new about our users' needs, it was just information we already knew presented in a different way. This is the same for the CTA, it was nice breaking down the steps, however, it was very straightforward and did not give me new information. Since they just repeated old information, these methods felt like a waste of time and didn't give me a fresh direction for the design.

b. How useful was the well-established questionnaire you selected (SUS or UEQ)?

The System Usability Scale (SUS) was a highly useful tool because it gave me a simple and reliable way to measure the overall user-friendliness of the application. After users interacted with our app, they completed the standard 10-question survey, which asked them to rate their agreement with statements like "I found the system unnecessarily complex" on a simple scale. The great strength of the SUS is that it takes these individual answers and calculates a single score from 0 to 100. This number was incredibly valuable because it provided a clear snapshot of our app's usability health, allowing us to track our progress over time and, most importantly, to benchmark our score against industry standards to see if our app was considered "good," "excellent," or "poor" compared to others.

However, the SUS was excellent for answering the question "How usable is it?", it was not at all useful for answering the critical question of "Why?" The SUS score gave us the insight of the issues that the users are facing in a general way, however, it didn't tell us the details and the specifics. For instance, our score might have been low, but the SUS couldn't pinpoint whether the problem was a confusing setup process, slow performance, or a fear of sending photos to the wrong person. To get those specific, actionable insights, we had to rely on other methods like user interviews and observation. The interviews allowed us to have a conversation and dig into the "why" behind the scores, uncovering the emotional frustrations and specific confusing steps that the SUS could only hint at through a low number. So, the SUS was a fantastic tool for measurement and justification, but it was not a tool for diagnosis or solution-finding.